Group Name: Mary and her little lamb's

Section: BSIS – 2A

Part 3: NoSQL Reflection Activity

Inventory and Sales Management System

In developing the Inventory and Sales Management System, the primary goal was to create a database-driven application capable of efficiently handling over 100,000 transaction records. While the current relational schema successfully manages core entities like suppliers, customers, and products, the Inventory Control and High-Volume Reporting components would benefit significantly from a document-based NoSQL approach.

Currently, the inventory log table tracks basic stock movements. However, in a large-scale retail environment, the system must also capture "noisy" unstructured data—such as detailed system audit trails, raw JSON payloads from API errors, and high-frequency clickstream data from the Cashier or Inventory Clerk. Storing these logs in a NoSQL document store prevents the primary SQL database from being bogged down by non-transactional data, ensuring that Transaction Processing remains rapid and consistent.

Furthermore, the products table currently uses a fixed structure for shoe details like price and size. A NoSQL approach would allow for more flexible Stock Ingestion of varied metadata, such as specific cushioning technologies or eco-friendly material percentages, without requiring a schema change.

Example NoSQL Document:

```
{
"product_id": 1001,
"name": "Adidas Ultraboost 22",
"sku": "ADI-UB22-WHT",
"metadata": {
"specs": {
"midsole": "Boost Foam",
"upper": "Primeknit",
"outsole": "Continental™ Rubber"
},
"tags": ["running", "cushioned", "breathable"]
}
}
```

By offloading these complex logs and metadata to a NoSQL system, the architecture maintains high performance for its High-Volume Reporting and analytics. This hybrid strategy ensures the system can provide real-time visibility into sales performance while remaining adaptable to the evolving data needs of a footwear business.